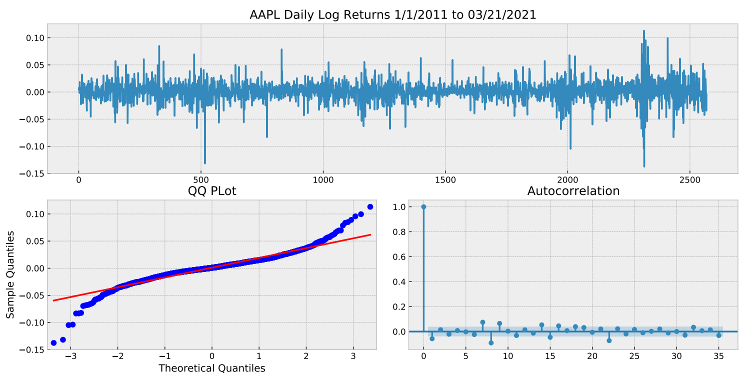
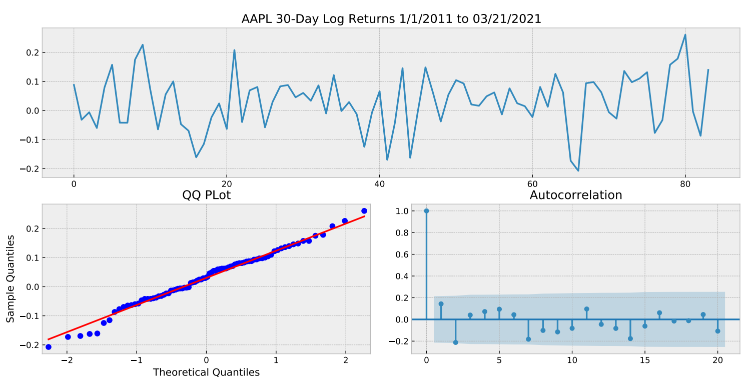
Financial Time Series
46-929
Solutions #1

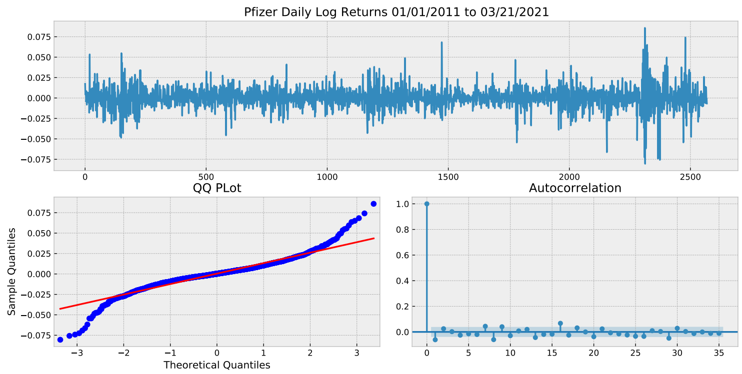
1) **Practice in creating time series plots**

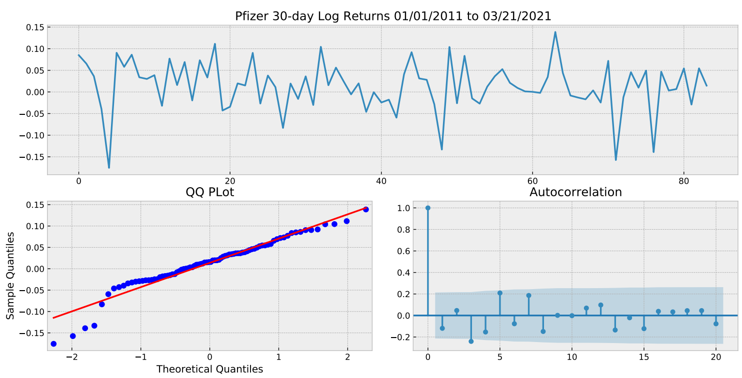
Imitating the graphs presented in class, pick any two of the four stocks: Apple (aapl), Pfizer (pfe), Gamestop (gme) and Bitcoin (btc-usd) and create plots for the log-returns, a q-q plot for the log-returns and an acf plot for the log returns. Do your plots give evidence for or against the stylized facts of Rama Cont?

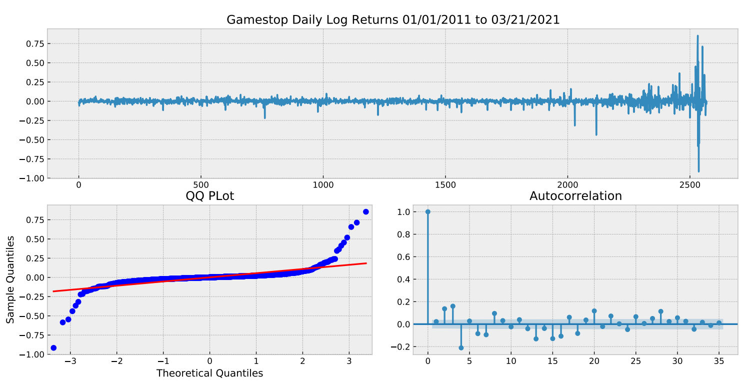
Using the code presented in lecture 1 including the function tsplot, we get the following plots for the four examples listed above:

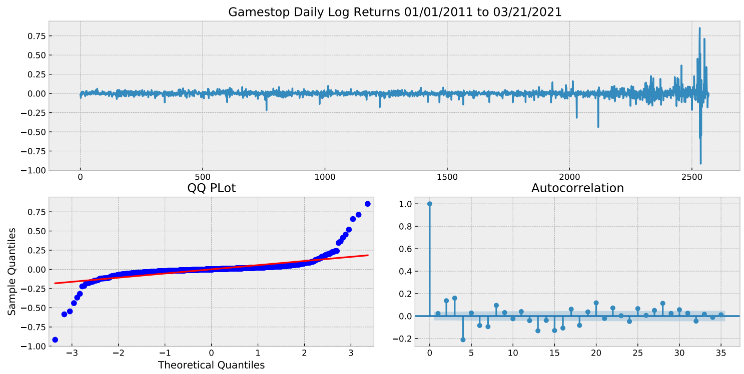


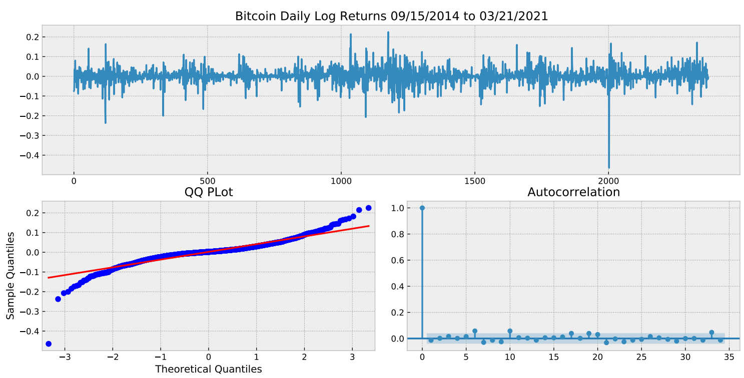


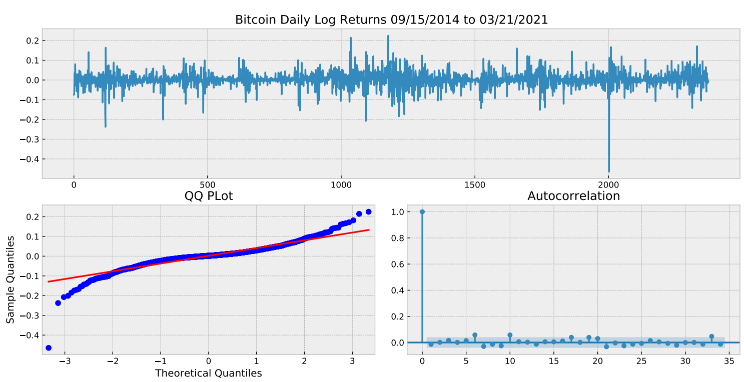

Pfizer Daily Log Returns 01/01/2011 to 03/21/2021
QQ PLot
Sample Quantiles
Theoretical Quantiles
Autocorrelation


Pfizer 30-day Log Returns 01/01/2011 to 03/21/2021
QQ PLot
Sample Quantiles
Theoretical Quantiles
Autocorrelation


Gamestop Daily Log Returns 01/01/2011 to 03/21/2021
QQ PLot
Sample Quantiles
Theoretical Quantiles
Autocorrelation



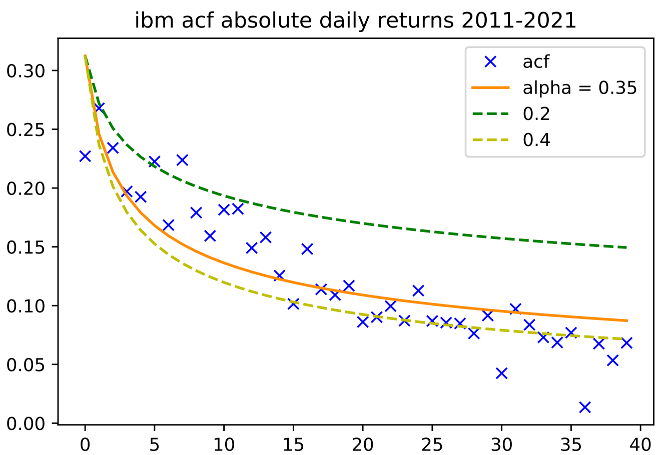


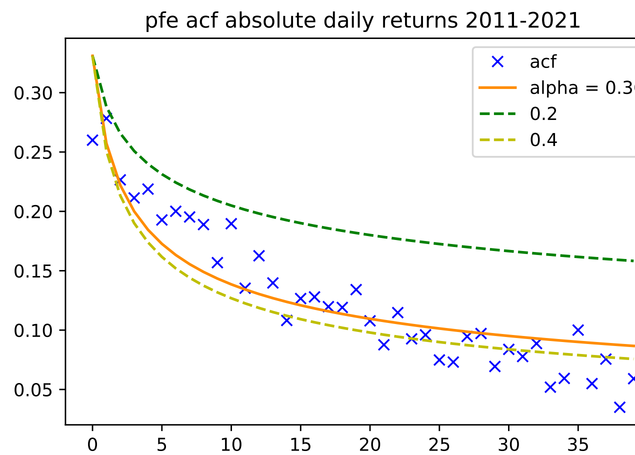


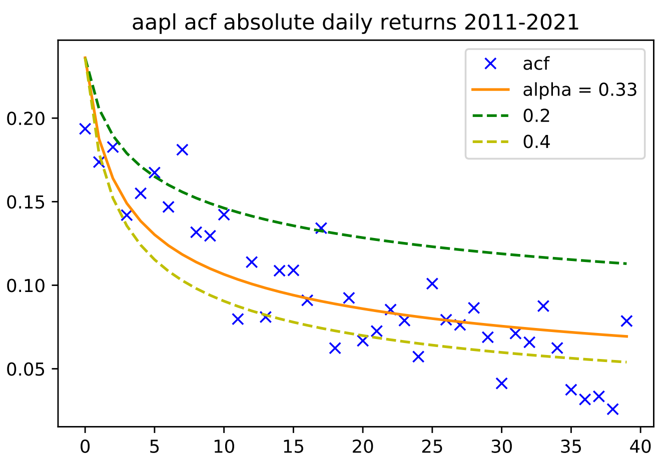
2) **Behavior of absolute returns**
One of the stylized principles of Rama Cont is the slow decay of absolute returns: "the autocorrelation funciton of absolute returns decays slowly as a function of the time lag, roughly as a power law with an exponent $\beta \in [0.2, 0.4]$." Explore this by taking a few stocks of your choice and seeing if they follow this stylized fact.

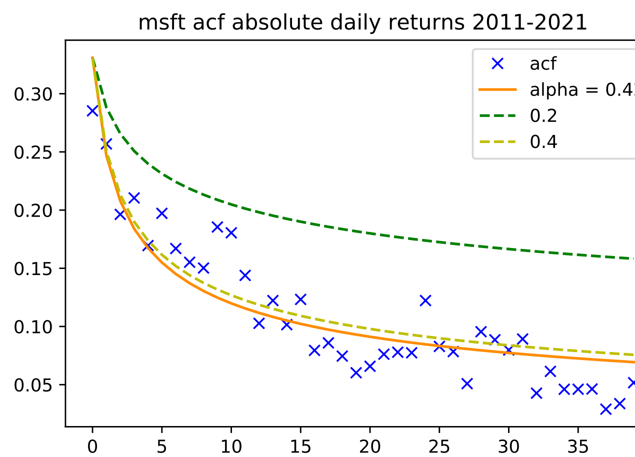
Below are 4 figures for IBM, Pfizer, Apple and Microsoft for the absolute value of daily returns over the period 1/1/2011 to 3/21/2021. According to Rama Cont, these should decrease as a power function $f(x) = c/x * *\alpha$ with exponent between [.2,.4]. The power function is fit using curve_fit from scipy.optimize where the fitting is done with two parameters, c and α.

We find values for α in these four cases to be IBM (.35), Pfizer (.36), Apple (.33) and Microsoft (.42), with only msft falling outside the Rama Cont region. the Rama Cont conjectured interval. The graphs below show the acf of the absolute daily returns, the fitted power function, and the power functions with $\alpha = .2$ and .4. The Rama Cont stylized fact seems plausible, although it is not clear that the power function is really the best description of the behavior of the acf. Still it does serve as a useful, if rough, description of this.









```
import numpy as np
import pandas as pd
import pandas_datareader as pdr
import statsmodels as smm
import statsmodels.formula.api as smf
import statsmodels.tsa.api as smt
import statsmodels.api as sm
from statsmodels.tsa.stattools import acf
import scipy
import scipy.stats as scs
from scipy.optimize import curve_fit
import matplotlib.pyplot as plt
import matplotlib as mp
import seaborn as sns

def power_fcn(x,alpha,c):
    return c/(x**alpha)

start = '2011-01-02'
end = '2021-03-21'
name = 'pfe'
stk = pdr.get_data_yahoo(name,start,end)['Adj Close']

stk_ret = np.log(stk).diff().dropna()
stk_absret = np.absolute(stk_ret)
stk_acf = acf(stk_absret)[1:]

x = np.arange(1,len(stk_acf)+1)
params = curve_fit(power_fcn,x,stk_acf)
alpha, c = params[0][0], params[0][1]
print('Alpha =',alpha)
plt.plot(stk_acf,'bx')
plt.plot(power_fcn(x,alpha,c),'darkorange')
plt.plot(power_fcn(x,0.2,c),'g--')
plt.plot(power_fcn(x,0.4,c),'y--')
plt.legend(['acf','alpha = ' + str(round(alpha,2)),'0.2','0.4'])

plt.title(name + ' acf absolute daily returns 2011-2021') plt.savefig(name + '.pdf')
plt.show()
```

3) Let $\{\epsilon_t, \ -\infty < t < \infty\}$ be a sequence of independent random variables with common mean 0 and common variance σ^2. For each of the three processes $\{X_t, \ -\infty < t < \infty\}$ below, which are weakly stationary processes? For those that are, find the mean and covariance function.

   a) $X_t = a_0 + a_1\epsilon_t + a_2\epsilon_{t-1}$ for constants $a_i, \ i = 1, 2, 3$.

   b) $X_t = a_0 + a_1 f(\epsilon_t)$ for constants a_0, a_1 and some function f.

c) $X_t = \Pi_{i=0}^{k} \epsilon_{t-i},\ k \geq 1.$

a) The process is weakly stationary

i. $E(X_t) = a_0 + a_1 E(\epsilon_t) + a_2 E(\epsilon_{t-1}) = a_0 + a_1 \cdot 0 + a_2 \cdot 0 = a_0$, so the process has constant mean 0.

ii.

$$\begin{aligned} Cov(X_t, X_{t+h}) &= Cov(a_0 + a_1\epsilon_t + a_2\epsilon_{t-1}, a_0 + a_1\epsilon_{t+h} + a_2\epsilon_{t-1+h}) \\ &= a_1^2 Cov(\epsilon_t, \epsilon_{t+h}) + a_1 a_2 Cov(\epsilon_t, \epsilon_{t-1+h}) + a_1 a_2 Cov(\epsilon_{t-1}, \epsilon_{t+h}) \\ &\quad + a_2^2 Cov(\epsilon_{t-1}, \epsilon_{t+h-1}) \end{aligned}$$

Since the ϵ_t sequence is independent, all any of the covariance terms above is 0 unless the time indicies are identical, in which case it is equal to σ^2. If $h = 0$, then the first and fourth covariances are σ^2, the other two are 0. If $|h| = 1$, then the second term is σ^2 while the other three are 0. In all other cases, all covariance terms are 0. To summarize:

$$\gamma(h) = \begin{cases} (a_1^2 + a_2^2)\sigma^2 & h = 0. \\ a_1 a_2 \sigma^2 & |h| = 1, \\ 0 & |h| > 1. \end{cases}$$

b) Since the ϵ_t sequence is independent but not identically distributed, the X_t sequence is not, in general, stationary. If f is a linear function, then this would be a special case of (a) and, therefore, weakly stationary. But even if the first and second moments of the ϵ_t sequence are identical, it does not follow that the moment of the $f(\epsilon_t)$ sequence are identical. Moreover, there is no guarantee that the first two moments of $f(\epsilon_t)$ exist let alone be equal $\forall t$.

c) This process is weakly stationary. By independence

$$\begin{aligned} E(X_t) &= E(\Pi_{i=0}^{k} \epsilon_{t-i}) \\ &= \Pi_{i=0}^{k} E(\epsilon_{t-i}) \\ &= 0 \end{aligned}$$

$Cov(X_t, X_{t+h}) = Cov(\Pi_{i=0}^{k} \epsilon_{t-i}, \Pi_{i=0}^{k} \epsilon_{t+h-i}) = E(\Pi_{i=0}^{k} \epsilon_{t-i} \cdot \Pi_{i=0}^{k} \epsilon_{t+h-i}) - 0 \cdot 0$. If $h = 0$ then the result (again by independence) is σ^{2k}. If $h \neq 0$, then at least two of the ϵ terms in the previous expectation are to the first power. Applying independence, their expectation is 0, hence the covariance is 0. Thus the X_t process is white noise with covariance function:

$$\gamma(h) = \begin{cases} \sigma^{2k+2} & \text{if } h = 0, \\ 0 & \text{if } |h| \neq 0. \end{cases}$$

4 Let $\{\epsilon_t,\ -\infty < t < \infty\}$ be an independent sequence with common mean 0 and common variance σ^2. Let

$$X_t = \epsilon_t + \epsilon_{t-1}\epsilon_{t-2}.$$

Determine if $\{X_t,\ -\infty < t < \infty\}$ is a white noise process and/or a MDS.

The process is weakly stationary but not an MDS. $E(X_t) = E(\epsilon_t + \epsilon_{t-1}\epsilon_{t-2}) = 0 + 0 \cdot 0$, but independence.

$$\begin{aligned} Cov(X_t, X_{t+h}) &= Cov(\epsilon_t + \epsilon_{t-1}\epsilon_{t-2}, \epsilon_{t+h} + \epsilon_{t+h-1}\epsilon_{t+h-2}) \\ &= Cov(\epsilon_t, \epsilon_{t+h}) + Cov(\epsilon_{t-1}\epsilon_{t-2}, \epsilon_{t+h}) \\ &+ Cov(\epsilon_t, \epsilon_{t+h-1}\epsilon_{t+h-2}) + Cov(\epsilon_{t-1}\epsilon_{t-2}, \epsilon_{t+h-1}\epsilon_{t+h-2}) \end{aligned}$$

Hence the autocovariance function is given by

$$\gamma(h) = \begin{cases} \sigma^2 + \sigma^4 & \text{if } h = 0, \\ 0 & \text{if } |h| \neq 0, \end{cases}$$

since the second and third terms are 0 while the first and fourth are σ^2 and σ^4 respectively. Thus the X_t process is white noise.

The process is not an MDS.

$$\begin{aligned} E(X_{t+1}|\mathcal{F}_t) &= E(\epsilon_{t+1} + \epsilon_t\epsilon_{t-1}|\mathcal{F}_t) \\ &= 0 + \epsilon_t\epsilon_{t-1}. \end{aligned}$$

Consequently, $E(X_{t+1}|\mathcal{F}_t) = \epsilon_t\epsilon_{t-1} \neq 0$, hence X_t is not an MDS.